

article and space limitation, it is not possible to entirely cover the APT image format, but when receiving these images, it is equally important to understand the image data properly. Therefore, this section has been limited to the basic information only. Readers are advised to refer to

the NOAA website for further information.

Receiving and decoding weather images

The APT offers us a resolution of 4km/pixel.

Also, there is the Meteor M2 satellite from Russia that transmits the earth imagery in the same 137MHz weather band, currently at 137.1MHz but in digital format at 72kbps QPSK. It offers even higher resolution of 1km/pixel in stunning three spectral bands as against only two channels from the NOAA APT.

Therefore, the decoding software WXtoImag used in the first article is unsuitable for this satellite. We require both the QPSK demodulator and the image extractor from the demodulated data.

The imager in Meteor is called MSU-MR or MultiSpectral Scanner for meteorological purposes.

Table 2 summarises the spectral bands comparison of NOAA and Meteor M2 satellites.

Software options

Since this satellite offers higher resolution of 1km/pixel, let us now see how to demodulate the signal and

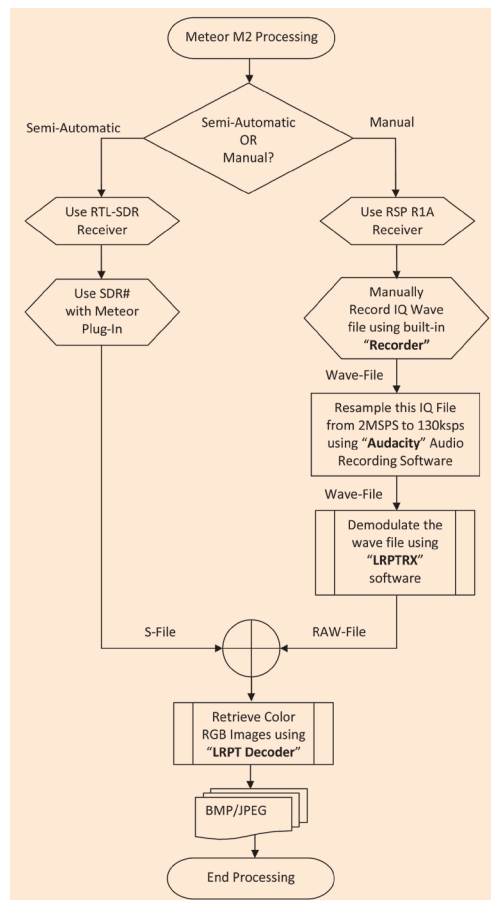


Fig. 2: The flow-chart showing the demodulation and image retrieval process

TABLE 2

SPECTRAL BANDS COMPARISON OF NOAA AND METEOR M2 SATELLITES

NOAA-1x Polar Satellites		Russian Meteor M2 Satellite	
Channel No.	Wavelength range	Channel No.	Wavelength range
1	580nm – 680nm	1	500nm – 700nm
2	725nm – 1000nm	2	700nm – 1100nm
3A	1580nm – 1640nm	3	1600nm – 800nm
3B	3.55μm – 3.93μm	4	3.5μm – 4.1μm
4	10.20 μm – 11.30μm	5	10.5μm – 11.5μm
5	11.5 μm – 12.5μm	6	11.5μm – 12.5μm

Note: The Meteor M2 spectral information was obtained from the website of Planeta (Scientific Research Center for Space HydroMeteorology, Russia)

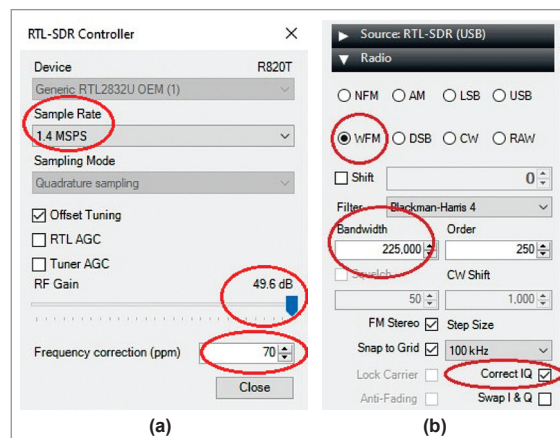


Fig. 3: (a) SDR settings, (b) Receiver settings

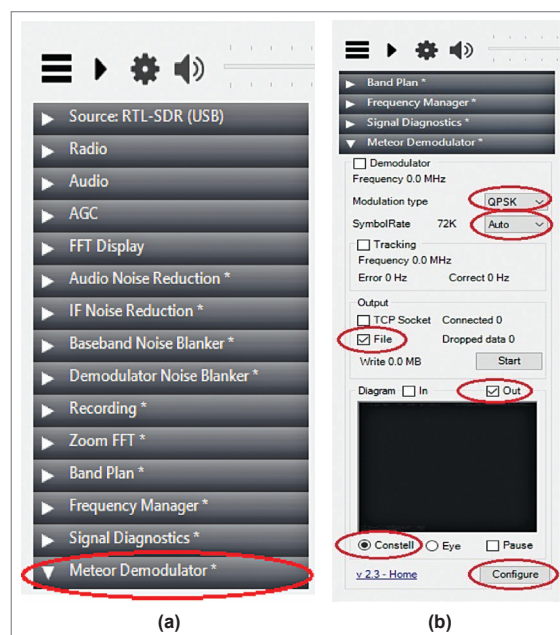


Fig. 4: (a) Properly installed plug-in, (b) Demodulator settings

obtain the earth imagery.

There are two options, and both require manual processing or file handling. Nothing gets automatically processed like the analogue APT software that we use for NOAA satellites. The options are:

Option 1: SDR# software with built-in Meteor M2 demodulator

Option 2: Manually recording the SDR I/Q data in a file and then processing it

Both these options give a demodulated output file, which needs to be processed manually to get the RGB colour/spectral images from the